

Mini Project Presentation

Subject Code:

Topic: 🇮🇳 Stock Market Analysis Using Simple Moving Average Strategy with ML

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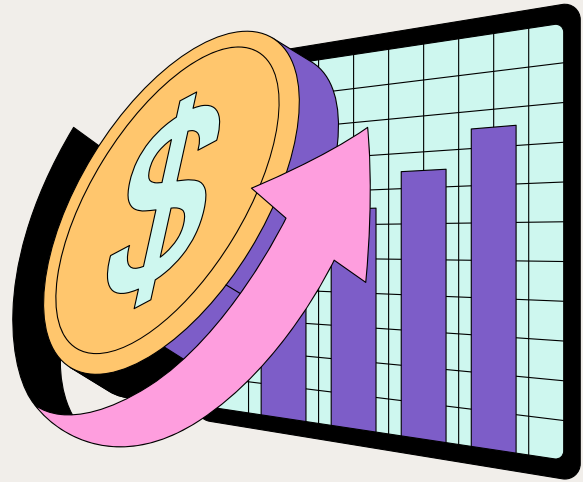
my mini project on Stock Market Analysis Using Simple Moving Average Strategy with Machine Learning. This project blends both theory and practical implementation, with a focus on financial data analysis and technical indicators



Mini Project

Stock Market Analysis Using SMA Strategy

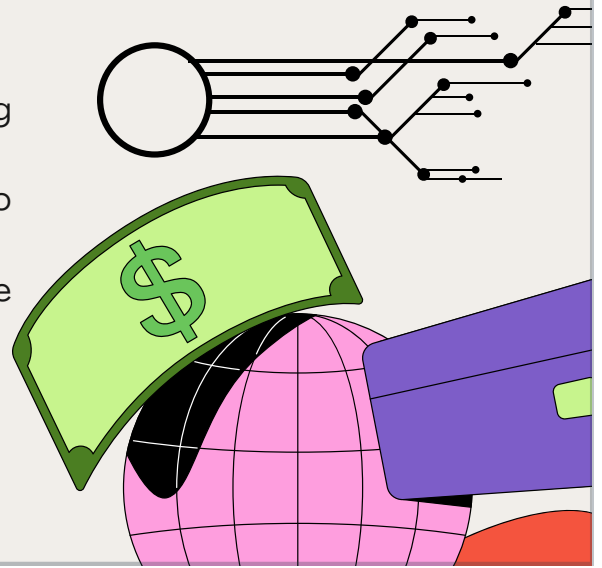
A Theoretical and Application-Based Approach



"This project is both theoretical and application-based. It explores how technical indicators like the Simple Moving Average, or SMA, can be used to design trading strategies. Additionally, it looks at how Machine Learning can further enhance these strategies for real-world usage."

Objective

- To implement and evaluate a simple trading strategy based on moving averages.
- Analyze historical stock data (TCS) to generate buy/sell signals using SMA.
- Understand how technical indicators guide investment decisions.



"The objective of this project was to implement a trading strategy using SMA. I used historical stock data for TCS to generate buy and sell signals based on SMA values. The broader goal was to understand how such indicators influence investment decisions in practice."

Theoretical Background

What is SMA?

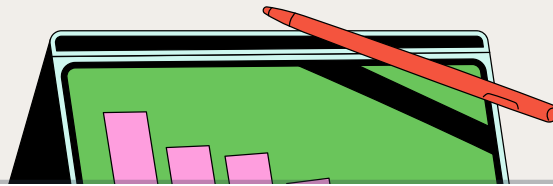
- SMA is the average of a selected range of prices, typically closing prices, by the number of periods in that range.

Why SMA?

- Helps smooth out price data.
- Identifies trends over time.

Trading Strategy

- Buy Signal: Short-term SMA crosses above long-term SMA.
- Sell Signal: Short-term SMA crosses below long-term SMA.



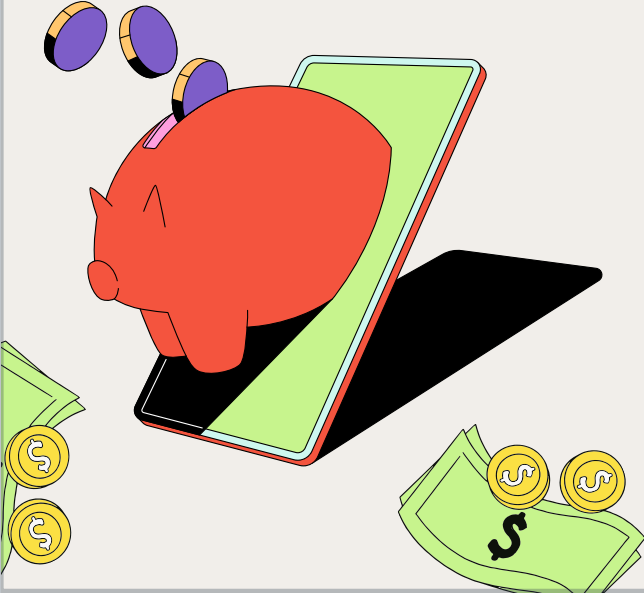
"Let me explain the core concept. The Simple Moving Average is a statistical measure used to smooth out price data over a period. Traders often use two different SMAs — a short-term and a long-term — to identify trends. A buy signal occurs when the short-term SMA crosses above the long-term one, and a sell signal when it crosses below."

Dataset

- Stock: TCS (Ticker: TCS.NS)
- Source: Historical Data of TCS stock prices from finance.yahoo.com for year of **2022 to 2024**
- Columns: Date, Open, High, Low, Close, Adj Close, Volume



"For this project, I used stock data of TCS from Yahoo Finance, covering the years 2022 to 2024. The dataset includes common stock market parameters like opening price, closing price, and volume. This data provides a strong foundation for calculating our indicators."



Methodology

- Import data using pandas.
- Convert date and set as index.

Compute:

- 20-day SMA
- 50-day SMA

• Generate signals:

- +1 for buy
- -1 for sell

"The first step was to import and preprocess the data using Python and Pandas. I then calculated the 20-day and 50-day SMAs. Buy and sell signals were generated based on the crossover strategy: +1 for buy and -1 for sell. These signals form the core of our strategy simulation."

Application

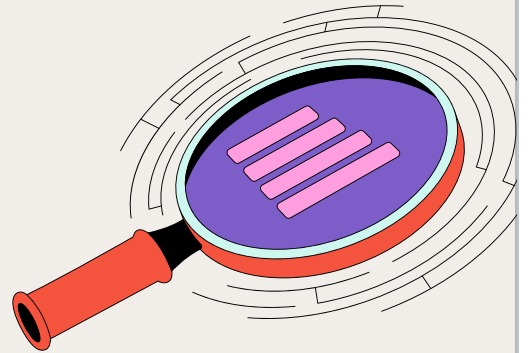


- This strategy can be used by:
 - Retail investors for simplified decision-making.
 - Financial analysts to prototype trading bots.
- Can be scaled to include other indicators like RSI, MACD, Bollinger Bands.

"This kind of strategy can help retail investors make informed decisions with minimal effort. Financial analysts can use this as a prototype for trading bots. It also provides a base that can be extended using more advanced indicators like RSI, MACD, and Bollinger Bands."

- SMA reacts slowly to market changes.
- May give false signals in sideways markets.
- Doesn't account for external news or market sentiment.

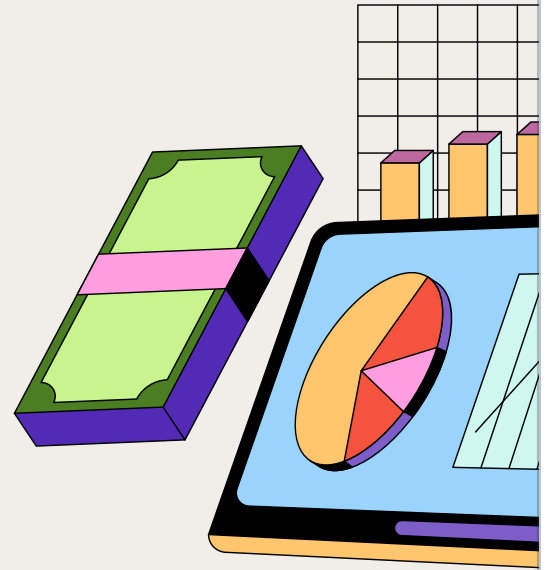
Limitations



"While SMA is useful, it has its drawbacks. It's slow to react to market changes and can give false signals in sideways markets. It also doesn't account for qualitative factors like market sentiment or external news."

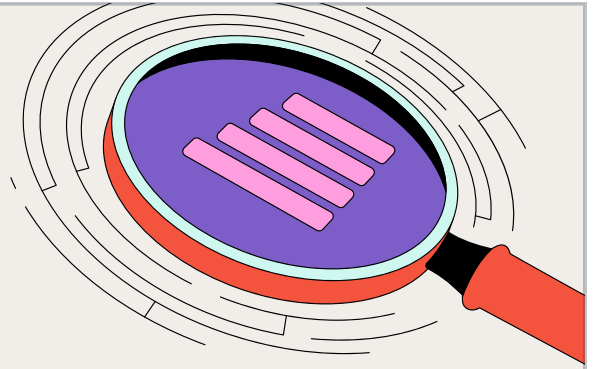
Future Work

- Incorporate more advanced techniques like:
 - Exponential Moving Averages (EMA)
 - Machine Learning (e.g., LSTM, Random Forest)
- Backtesting using multiple stock symbols
- Visualization dashboards



"To improve this system, we can use exponential moving averages or apply machine learning models like LSTM and Random Forests. We can also backtest strategies on multiple stocks and develop a dashboard for better visualization."

The Role of Machine Learning in Stock Analysis



Why ML?

Traditional strategies like SMA are static and may not capture market complexities. ML can model non-linear relationships and hidden patterns.

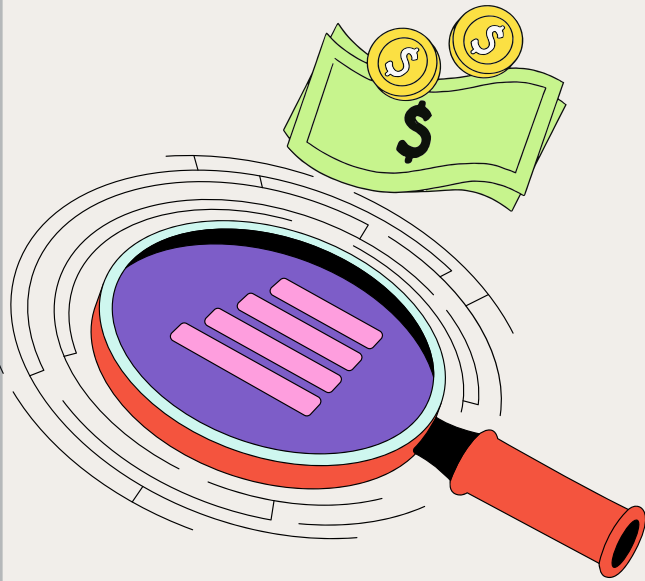
Next Step:

Use SMA signals as features for training ML models – blending rule-based logic with data-driven learning.

Advantages of ML Approaches:

- Adaptability to changing market conditions
- Ability to process large, multidimensional datasets
- Potential for higher accuracy in complex scenarios

"Machine learning offers powerful tools for stock analysis. Unlike static SMA rules, ML can learn from large datasets and find hidden patterns. We can use SMA-based signals as input features and train models to make more adaptive predictions."



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- Machine Learning (ML) can complement SMA by learning from data patterns, enabling smarter and adaptive trading strategies.
- This project lays the foundation for integrating ML models like decision trees, random forests, or neural networks in future developments.

Conclusion

"To conclude, SMA provides a solid baseline for market trend analysis. But when combined with Machine Learning, we can build smarter, more flexible systems that better adapt to real-world market dynamics. This project is a first step toward such intelligent financial systems."

**THANK
YOU**